

SM SERVICES - INVOICE & BILLING SYSTEM

Comprehensive Technical & System Documentation | Enterprise Release 2026

1. Executive Overview

The **SM SERVICES Invoice & Billing Automation System** is a full-stack, enterprise-grade web application designed to streamline billing operations, receipt data extractions, and financial reporting. Built with a high-performance Flask architecture and styled in a customized **Cyan Slate Neon UI Theme**, it delivers an intuitive interface alongside robust PDF generation capabilities.

2. Technology Stack & Dependencies

Component	Technology / Library	Purpose & Role
Backend Framework	Flask 3.x (Python)	Application routing, session management, RESTful endpoints.
Database	SQLite3	Relational data storage for invoices and user credentials.
Authentication	Flask-Login & Werkzeug	Session persistence, route security, password hashing.
PDF Engine	ReportLab Engine	Dynamic generation of corporate-styled PDF tax invoices.
OCR Processing	Pytesseract & PIL	Extracting text data from uploaded physical receipt images.
Document Scanning	pdfplumber & python-docx	Scanning digital PDF and Word documents for data pre-fill.
Payment Integration	qrcode Engine	Generating dynamic UPI payment QR codes on-the-fly.
Frontend UI / UX	Bootstrap 5 & Custom CSS3	Dark Slate Slate/Neon responsive user interface.

3. Core System Features

- **Authentication & Access Control:** Secured user sessions via Flask-Login. Passwords stored using PBKDF2 hashing with Werkzeug security tools.
- **Automated Invoice Counter:** Sequential, structured invoice number generation formatted as: YYYY-YY/SM/DIC/XX.
- **Multi-Format OCR Scanner:** Extracts structured text from receipt images (.png, .jpg), PDFs, and Word documents (.docx).
- **Dynamic Itemization Table:** Interactive DOM manipulation allowing users to add/delete line items with automatic tax calculations.
- **Dynamic UPI QR Generator:** Auto-embeds merchant UPI payment QR codes directly into generated PDF receipts.
- **Dark Slate Neon UI Design:** Modern, high-contrast dark theme utilizing #0B132B (Slate) and #00F5D4 (Neon Cyan) palette.
- **Real-time Sales Dashboard:** Track total revenue generated, pending invoice counts, and instant live table search filtering.

4. Route Map & Application Endpoints

Endpoint	HTTP Method	Auth Required	Description
/	GET	No	Public Landing / Marketing Page.
/login	GET / POST	No	User authentication & credential validation.
/logout	GET	Yes	Terminates active session and clears cookies.
/app	GET	Yes	Main Invoice Generator workspace UI.

/scan_receipt	POST	Yes	OCR receipt image & file scanner API.
/generate_pdf	POST	Yes	Builds PDF invoice and updates database.
/dashboard	GET	Yes	Analytics dashboard and invoice database.

5. Database Architecture (database.db)

Table: users

- **id:** INTEGER (Primary Key, Auto-increment)
- **username:** TEXT (Unique, Not Null)
- **password:** TEXT (Hashed Password String)

Table: invoices

- **id:** INTEGER (Primary Key)
- **invoice_no:** TEXT (Unique)
- **date / supplier_name / client_name:** TEXT
- **gross_amount / tax_amount / total_amount:** REAL
- **status:** TEXT (Default: 'Pending')

6. Setup & Execution Guide

1. **Install Dependencies:** `pip install flask flask-login py-tesseract pdfplumber python-docx qrcode reportlab pillow werkzeug`
2. **Install OCR Engine:** Install Tesseract OCR on system path for receipt image scanning capabilities.
3. **Initialize & Run App:** Run `python app.py`. The app automatically provisions database.db and default credentials (**admin / admin123**).